

RES. 6-012 Introduction to Probability - Lectures 16
Textbook Problems - Sections 8.3 & 8.4

Problem Set - Section 8.3

- 10) Radar overestimates cars Uniform $\sim (0, 5)$ miles/hour
Car speed Uniform $\sim (55, 75)$ miles/hour

~~X~~ $X \rightarrow$ Radar's measurement $= \Theta + W$
 ~~Θ~~ $\Theta \rightarrow$ car speed $= \Theta$

Prior: ~~$f_{\Theta}(\theta)$~~ $f_{\Theta}(\theta) = \frac{1}{(75)-(55)} = \frac{1}{20}$; $\theta \in [55, 75]$

Observation: $f_{X|\Theta}(x|\theta) = \frac{1}{(5+\theta)-(0+\theta)} = \frac{1}{5}$ for $x \in [0, 5]$
 $x \in [\theta, \theta+5]$

Posterior: $f_{\Theta|X}(\theta|x) = \frac{f_{\Theta}(\theta) \cdot f_{X|\Theta}(x|\theta)}{\int f_{\Theta}(\theta') \cdot f_{X|\Theta}(x|\theta') d\theta'}$

~~$f_{\Theta|X}(\theta|x) = \frac{(\frac{1}{20})(\frac{1}{5})}{\int_{55}^{75} f_{\Theta}(\theta') f_{X|\Theta}(x|\theta') d\theta'}$~~

$f_{\Theta|X}(\theta|x) = \frac{(\frac{1}{20})(\frac{1}{5})}{\int_{55}^{75} f_{\Theta}(\theta') f_{X|\Theta}(x|\theta') d\theta'}$

Bounds: prior: $\theta' \in [55, 75]$ prior = 0 outside

likelihood: ~~$\theta' \in [0, x]$~~ $x \in [\theta', \theta'+5]$

$\theta' \leq x \leq \theta'+5$

$\theta' - x \leq 0 \leq \theta' + 5 - x$

$-x \leq -\theta' \leq 5-x$

x full range: $x \in [55, 80]$ $x-5 \leq \theta' \leq x$

$\theta' \in [\max(55, x-5) ; \min(75, x)]$

$\therefore \theta' \in [x-5 ; x]$

$$\begin{aligned}
 f_{\theta|x}(\theta|x) &= \frac{\left(\frac{1}{100}\right)}{\int_{x-5}^x \left(\frac{1}{20}\right) \left(\frac{1}{5}\right) d\theta'} \\
 &= \frac{\left(\frac{1}{100}\right)}{\frac{1}{100} \left(\theta'\right) \Big|_{x-5}^x} \\
 &= \frac{1}{x - (x-5)} \\
 &= \frac{1}{5} \rightarrow
 \end{aligned}$$

LMS Estimator posterior Uniform $[x-5, x]$

$$E[\theta | X = x] = \cancel{\frac{1}{(x) - (x-5)} = \frac{1}{5}} \quad \left(\frac{a+b}{2}\right)$$

$$= \frac{(x-5) + (x)}{2}$$

$$= \frac{2x-5}{2}$$

$$= x - 5/2$$

↳ Only 1 case handled!!

12) $\theta = \theta$ $X_1, \dots, X_n \rightarrow \text{indep ; uniform } [0, \theta]$

Prior dist: θ uniform $[0, 1]$ assume $n > 3$.

a) $LMS = E[\theta | X_1, \dots, X_n]$

Prior $f_\theta(\theta) = \frac{1}{b-a}$
 $= \frac{1}{1-0}$ $0 \leq \theta \leq 1$
 $f_\theta(\theta) = 1$

Observations $f_{x|\theta}(x|\theta) = \prod_{i=1}^n f_{x_i|\theta}(x_i|\theta)$

$f_{x|\theta}(x|\theta) = 0$ if $x_i > \theta$ }
 $\therefore \theta \geq \max\{x_1, \dots, x_n\}$ }
 $\text{Let } \bar{x} = \max\{x_1, \dots, x_n\}$ }
 $= \prod_{i=1}^n \frac{1}{\theta - 0}$; $0 \leq x_i \leq \theta$
 $= \frac{1}{\theta^n}$ $\therefore \theta \in [\bar{x}, 1]$

Posterior $f_{\theta|X_1, \dots, X_n}(\theta | x_1, \dots, x_n) =$

$= \frac{f_\theta(\theta) \cdot f_{x|\theta}(x|\theta)}{\int f_\theta(\theta') f_{x|\theta}(x|\theta') d\theta'}$

$= \frac{(1) \left(\frac{1}{\theta^n} \right)}{\int_{\bar{x}}^1 (1) \left(\frac{1}{\theta'^n} \right) d\theta'}$

$= \frac{\frac{1}{\theta^n}}{\int_{\bar{x}}^1 \frac{1}{\theta'^n} d\theta'} = \frac{\frac{1}{\theta^n}}{\int (\theta')^{-n} d\theta'}$

$= \frac{\frac{1}{\theta^n}}{\left(\frac{(\theta')^{-n+1}}{-n+1} \right) \Big|_{\bar{x}}^1} = \frac{\frac{1}{\theta^n}}{\left(\frac{1^{-n+1}}{-n+1} \right) - \left(\frac{\bar{x}^{-n+1}}{-n+1} \right)}$

Walk through

Posterior $f_{\theta|x_1, \dots, x_n} = \begin{cases} \frac{\frac{1}{\theta^n}}{\frac{1}{n-1} \bar{x}^{1-n} - \frac{1}{n-1}} & ; \text{ if } \bar{x} \leq \theta \\ 0 & ; \text{ otherwise} \end{cases}$

LMS = $E[\theta | x_1, \dots, x_n] =$

WT $\left\{ \begin{aligned} & E[\theta | X = \bar{x}] = \int \theta \cdot f_{\theta|X}(\theta | \bar{x}) d\theta \\ & = E[\theta | x_1, \dots, x_n] = \int \theta \cdot f_{\theta|x_1, \dots, x_n}(\theta | x_1, \dots, x_n) d\theta \\ & = \int_{\bar{x}}^1 \theta \cdot \frac{\frac{1}{\theta^n}}{\frac{1}{n-1} \bar{x}^{1-n} - \frac{1}{n-1}} d\theta \quad \text{Let } c = \frac{1}{n-1} \bar{x}^{1-n} - \frac{1}{n-1} \\ & = \frac{1}{c} \int_{\bar{x}}^1 \theta^{-n+1} d\theta \\ & = \frac{1}{c} \cdot \left(\frac{\theta^{-n+2}}{-n+2} \Big|_{\bar{x}}^1 \right) = \frac{1}{c} \cdot \left(\frac{1^{-n+2}}{-n+2} - \frac{\bar{x}^{-n+2}}{-n+2} \right) \\ & = \frac{\frac{1}{n-2} - \frac{1}{n-2} \bar{x}^{-n+2}}{\frac{1}{n-1} \bar{x}^{1-n} - \frac{1}{n-1}} \quad (n > 3) \\ & = \frac{(\frac{1}{n-2}) (\bar{x}^{2-n} - 1)}{(\frac{1}{n-1}) (\bar{x}^{1-n} - 1)} \\ & = \frac{n-1}{n-2} \cdot \frac{\bar{x}^{2-n} - 1}{\bar{x}^{1-n} - 1} \end{aligned} \right.$

~~WT~~

c) $\bar{x} = 0.5 \rightarrow \bar{x}$ is a constant
MAP & LMS coincide. ✓

$\rightarrow$ MAP fixed at 0.5 (as max possible value)

$\rightarrow$ LMS determined by $\bar{x}$ and as such

$\hat{\theta} = [0.5, 1]$ will converge to 0.5 as $n \rightarrow \infty$ [0.5, 1]

$\rightarrow$ MSE as n grows, posterior concentrates around 0.5 (dist. squeezes a point) $\rightarrow$ MSE goes to zero

Section 8.4 Problem Sets

14) θ, x

LLMS.

$$E[\theta | x = x] = \int \theta \cdot f_{\theta|x}(\theta, x) d\theta$$

$$= \int \theta \cdot \frac{1}{12} d\theta$$

$$x = \theta + w$$

θ uniform $[4, 10]$
 w uni $[-1, 1]$

$$\hat{\theta} = E[\theta | x = x]$$

$$\hat{\theta} = E[\theta] + \frac{\text{cov}(\theta, x)}{\sigma_x^2} (x - E[x])$$

$$E[\theta] = \frac{4+10}{2} = 7 \quad E[x] = E[\theta + w] = E[\theta] + E[w]$$

$$E[w] = \frac{-1+1}{2} = 0 \quad \text{so } E[x] = (7) + (0) = 7$$

$$\sigma_x^2 = (\sigma_w)^2 + (\sigma_\theta)^2 \quad \sigma_w^2 = \frac{((-1) - (-1))^2}{12} = \frac{4}{12} = \frac{1}{3}$$

$$\sigma_\theta^2 = \frac{((10)^2 - (4)^2)}{12} = \frac{100}{12}$$

$$E[\theta] = E[\theta] + E[w] = E[x] = 7$$

$$= \frac{100 - 16}{12} = \frac{84}{12} = 7$$

$$\sigma_x^2 = 3 + \frac{1}{3}$$

$$\text{cov}(\theta, x) = E[x\theta] - E[x]E[\theta] =$$

$$= E[(x - E[x])(\theta - E[\theta])] =$$

$$= E[(\theta - E[x])^2] = \sigma_\theta^2$$

$$\hat{\theta} = (7) + \frac{(3)}{(3 + \frac{1}{3})} (x - (7))$$

$$= 7 + \frac{9}{10} (x - 7)$$

$$\text{MSE} : E[(\theta - \hat{\theta}_L)^2] = (1 - \rho^2) \sigma_\theta^2$$

$$\rho = \frac{\text{cov}(X, \theta)}{\sqrt{\sigma_x^2 \sigma_\theta^2}}$$

$$= \frac{\sigma_\theta^2}{\sigma_x \sigma_\theta} = \frac{\sigma_\theta}{\sigma_x}$$

$$\therefore \text{MSE} = \left(1 - \left(\frac{\sigma_\theta}{\sigma_x}\right)^2\right) \sigma_\theta^2$$

$$= \left(1 - \left(\frac{3}{3+1/3}\right)\right) \left(\frac{3}{2}\right)$$

$$= \left(1 - \frac{9}{10}\right) \left(\frac{3}{2}\right)$$

$$= \frac{3}{10} \checkmark$$

17) $\Theta \rightarrow$ positive RV

mean = μ

var = σ^2

Estimate $X = \sqrt{\Theta} \cdot W$

$W \rightarrow$ indep of Θ

$\hookrightarrow$ mean = 0

$\hookrightarrow$ var = 1

$\hookrightarrow E[W^4] \rightarrow$ known 4th moment

$$E[X | \Theta = \theta] = 0$$

$$\text{var}(X | \Theta = \theta) = \theta$$

$$\hat{\theta}_L = E[\Theta] + \frac{\text{cov}(\Theta, X)}{\text{var}(X)} (X - E[X])$$

$$E[X] = E[\sqrt{\Theta} \cdot W] = \cancel{E[\Theta]} \cdot \sqrt{E[\Theta]} \cdot E[W]$$

$$= 0$$

$$\text{var}(X) = E[(X - E[X])^2] = E[(X - 0)^2] = E[X^2]$$

$$\cancel{= E[\Theta] E[W^2]} = E[(\sqrt{\Theta} \cdot W)^2]$$

$$= E[\Theta \cdot W^2]$$

$$= E[\Theta] E[W^2]$$

$$\begin{aligned}\text{var}(W) &= 1 = E[W^2] - (E[W])^2 \\ 1 &= E[W^2] - (0)^2 \\ E[W^2] &= 1\end{aligned}$$

$$\begin{aligned}\circ \circ \text{var}(X) &= E[\Theta]E[W^2] \\ &= E[\Theta]\end{aligned}$$

$$\begin{aligned}\text{cov}(\Theta, X) &= \text{cov}(\Theta, \sqrt{\Theta}W) \\ &= \cancel{E[\Theta \sqrt{\Theta}W]} \\ &= \cancel{E[\Theta \sqrt{\Theta}W]} - E[\Theta]E[\sqrt{\Theta}W] \\ &= E[\Theta]\sqrt{E[\Theta]}E[W] - E[\Theta]\sqrt{E[\Theta]}E[W] \\ &= 0\end{aligned}$$

$$\begin{aligned}\hat{\Theta}_L &= E[\Theta] + \frac{\text{cov}(\Theta, X)}{\text{var}(X)} (X - E[X]) \\ &= \mu \quad \checkmark \quad \left(\frac{0}{E[\Theta]} \right) (X - 0) \\ &= \underline{\mu}\end{aligned}$$

$$\cancel{\Theta} Y = X^2 - (\sqrt{\Theta}W)^2 = \Theta W^2$$

$$E[\Theta] = \mu$$

$$E[X^2] = E[\Theta W^2] = E[\Theta]E[W^2] = E[\Theta] = \mu \quad \checkmark$$

$$\begin{aligned}\text{var}(X) &= \text{var}(\Theta W^2) = E[(\Theta W^2)^2] - E[\Theta] (E[\Theta W^2])^2 \\ &= E[\Theta^2 W^4] - \mu^2 \\ &= E[\Theta^2]E[W^4] - \mu^2 \\ &= (\text{var}(\Theta) + E[\Theta]^2)E[W^4] - \mu^2 \\ &= \sigma^2 + \mu^2 \quad \checkmark \\ &= (\sigma^2 + \mu^2)E[W^4] - \mu^2 \quad \checkmark\end{aligned}$$

$$\begin{aligned}\text{cov}(\Theta, X) &= \text{cov}(\Theta, \Theta W^2) = E[\Theta^2 W^2] - E[\Theta]E[\Theta W^2] \\ &= E[\Theta^2]E[W^2] - E[\Theta]^2E[W^2] \\ &= (\sigma^2 + \mu^2) - \mu^2 = \sigma^2 \quad \checkmark\end{aligned}$$

$$\Rightarrow \hat{\Theta}_L = \mu + \frac{\sigma^2}{(\sigma^2 + \mu^2)E[W^4] - \mu^2} (X^2 - \mu) \quad \checkmark$$

RES.6-012 Introduction to Probability - Lectures 17, 18
Textbook Problems - Sections 5.1, 5.2, 5.3.

Problem Set 5.1:

1) $M_n = \frac{(X_1 + \dots + X_n)}{n} \approx h$
 $\sigma_h = 1 \text{ m}$

a) $1 \text{ cm} = 0.001 \text{ m} \quad \sigma \leq 0.001 \text{ m}$

$$\sigma_{M_n} = \frac{1}{\sqrt{n}} \quad \text{var}(M_n) = \frac{\sigma^2}{n}$$
$$\sigma(M_n) = \frac{\sigma}{\sqrt{n}} = \frac{0.001}{\sqrt{n}}$$

$$0.001 \geq \frac{1}{\sqrt{n}}$$

$$0.001\sqrt{n} \geq 1$$

$$\sqrt{n} \geq 1000$$

$$n \geq 10,000$$

b)

$$P(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}$$

$$P = 0.99$$

$$5 \text{ cm} = 0.05 \text{ m}$$

$$0.99 \leq \frac{\sigma^2}{c^2}$$

$$0.99 \leq \frac{\sigma^2}{0.05^2}$$

$$\sigma^2 \geq 0.002475$$

$$0.99$$

$$\sigma \geq 0.049749$$

$$\therefore \sigma = \frac{1}{\sqrt{n}}$$

$$n = \left(\frac{1}{\sigma}\right)^2$$

$$= 404.0404$$

$$= 405$$

$$b) P(|M_n - h| \leq 0.05) \geq 0.99$$

$$\left\{ \begin{aligned} P(|M_n - h| \leq 0.05) &= 1 - P(|M_n - E[M_n]| \geq 0.05) \\ &\geq 1 - \frac{(\frac{1}{n})}{(0.05)^2} \end{aligned} \right\}$$

$$0.99 \geq 1 - \frac{(\frac{1}{n})}{(0.05)^2}$$

$$0.99(0.05)^2 \geq (0.05)^2 - (\frac{1}{n})$$

$$(\frac{1}{n}) \geq \frac{1}{40000}$$

$$\therefore n \geq 40000$$

$$c) h \in [1.4; 2.0]$$

$$\sigma^2 = \frac{(b-a)(b-a+2)}{12}$$

$$= \frac{(0.6)(2.6)}{12}$$

$$= \frac{13}{100}$$

$$\frac{1}{n} = 0.13$$

$$n = \frac{100}{13} = 7.692$$

$$\approx 8$$

X

Eg 5.3

$$\sigma^2 \leq (b-a)^2/4$$

$$\leq (2.0 - 1.4)^2/4$$

$$\leq (0.6)^2/4$$

$$\leq 0.09$$

$$\sigma \leq 0.3$$

$$\therefore (a) \quad n = \left(\frac{b-a}{\sigma}\right)^2 = \left(\frac{0.3}{0.03}\right)^2 = 900$$

$$\therefore (b) \quad 0.99 \geq 1 - \frac{(\frac{0.09}{n})}{(0.05)^2} \left\langle \left(\frac{1}{n}\right) = \left(\frac{0.3^2}{n}\right) \right\rangle$$

$$\left(\frac{0.09}{n}\right) \geq 0.01(0.05)^2 \geq \frac{1}{40000}$$

$$n \geq 3600$$

Problem Set 5.2

4) $n \rightarrow$ people $f \rightarrow$ frac of smokers

$$M_n = \frac{S_n}{n}$$

$$P(|M_n - f| \geq \epsilon) \leq \delta$$

$\underbrace{\hspace{2cm}}_{\text{tolerances}}$

a) $\epsilon \rightarrow \frac{1}{2}\epsilon$

$$P(|M_n - f| \geq \epsilon) \leq \delta$$

$$\frac{\sigma^2}{\epsilon^2} \leq \delta$$

$$\frac{\sigma^2}{(\frac{1}{2}\epsilon)^2} \leq \delta$$

$$\frac{\sigma^2}{\frac{1}{4}\epsilon^2}$$

$$4 \frac{\sigma^2}{\epsilon^2} \leq \delta$$

$$4 \frac{(\frac{1}{n})}{\epsilon^2} \leq \delta$$

$$\frac{1}{n} \leq \frac{1}{4}\epsilon^2\delta$$

$$n \leq \frac{4}{\epsilon^2\delta}$$

$\therefore n \rightarrow$ mult by 4.

b) $\frac{\sigma^2}{\epsilon^2} \leq (\frac{1}{2}\delta)$

$$\frac{(\frac{1}{n})}{\epsilon^2} \leq \frac{1}{2}\epsilon^2\delta$$

$$n \leq \frac{2}{\epsilon^2\delta}$$

$n \rightarrow$ mult by 2.

Problem Set 5.3

5) $X_1, X_2, \dots \rightarrow$ indep RVs (uniform $[-1, 1]$)
 $(-1 \leq X_n \leq 1)$

a) $Y_n = X_n/n$

$$\left\{ \begin{array}{l} \lim_{n \rightarrow \infty} P(|Y_n - a| \geq \epsilon) = 0 \\ \text{For any } \epsilon > 0: P(|Y_n| \geq \epsilon) = 0 \\ \text{for all } n \text{ with } \frac{1}{n} < \epsilon, P(|Y_n| \geq \epsilon) \rightarrow 0 \end{array} \right\}$$

$\rightarrow$ X bounded in $[-1, 1]$, so $\frac{X_n}{n} \leq 1/n$
 as $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$ } Key insight
 is bound from
 Uniform dist.

b) $Y_n = (X_n)^n$

$$(X_n)^n \leq (1)^n$$

$$(X_n)^n \leq 1 \Rightarrow [-1, 1], \text{ so } (X_n)^n \leq 1$$

$$\text{as } n \rightarrow \infty, (1)^n \rightarrow 1$$

$$P(|Y_n| \geq \epsilon) = P(|X_n|^n \geq \epsilon) = P(X_n \geq \epsilon^{1/n}) + P(X_n \leq -\epsilon^{1/n})$$

$$= 1 - \epsilon^{1/n}$$

as $n \rightarrow \infty$, $1 - \epsilon^{1/n} \rightarrow 1$ and $P(|Y_n| \geq \epsilon)$ approaches 0.

Cases

$|x| < 1 : x^n \rightarrow 0$

$x = 1 : 1^n \rightarrow 1$

$x = -1 : (-1)^n$ alternate ~~between~~ $\{-1, 1\}$.

$\rightarrow$ Prob any X_n exactly -1 or 1 ? $\rightarrow 0$

continuous RV at specific point $\leftarrow$

$P = 1.0$ % $|X_n| \leq 1$ $\leftarrow$ has no mass. $\infty X_n^n \rightarrow 0$

$$c) Y_n = X_1 \cdot X_2 \cdots X_n$$

$$P(|Y_n| \geq \epsilon) = P(|X_n| \geq \epsilon)$$

$X_1, X_2, \dots, X_n \rightarrow$ independent RVs

$$\therefore E[Y_n] = E[X_1] \cdots E[X_n] = 0$$

$$\underbrace{\underbrace{\mu=0} \cdots \underbrace{\mu=0}}_{=0}$$

symmetric around $[-1, +1]$

$$d) Y_n = \max\{X_1, \dots, X_n\}$$

$$P(|Y_n - 1| \geq \epsilon) = P(\max\{X_1, \dots, X_n\} \geq 1 + \epsilon) + P(\max\{X_1, \dots, X_n\} \leq 1 - \epsilon)$$

$$\stackrel{(2)}{=} P(X_1 \leq 1 - \epsilon, \dots, X_n \leq 1 - \epsilon)$$

(2) means max of all n values is below $1 - \epsilon$

$$= \left(1 - \frac{\epsilon}{2}\right)^n$$

(4) a number strictly < 1 raised to $n \rightarrow \infty$

$$\therefore P(|Y_n - 1| \geq \epsilon) \rightarrow 0$$

$$\therefore P(|Y_n - 1| \geq \epsilon) \rightarrow 0$$

(1) think if add more X 's $[-1, 1]$
 $\hookrightarrow$ value approaches $+1$ (max value)

$$Y_n = \max\{\dots\} \rightarrow 1 \text{ in probability.}$$

Problem Set 9.1

1) exponen distr θ $x = \{10, 14, 18, 8, 20\}$ $n=5$

$$E[X] = \frac{1}{\lambda} \quad f(x) = \lambda e^{-\lambda x} \quad ; x > 0.$$

$$E[X] = \frac{10 + 14 + 18 + 8 + 20}{5} = 14$$

$$\begin{aligned} f_X(x; \theta) &= f_{X_1}(x_1; \theta) \cdots f_{X_5}(x_5; \theta) \\ &= \theta e^{-x_1 \theta} \cdots \theta e^{-x_5 \theta} \\ &= \theta^5 e^{-(x_1 + \cdots + x_5) \theta} \\ &= \theta^5 e^{-(70) \theta} \\ &= \theta^5 e^{-70 \theta} \end{aligned}$$

Maximum Likelihood:

$$\hat{\theta}_{ML} = \theta^5 e^{-70 \theta}$$

$$\begin{aligned} \rightarrow \ell(\theta) &= \ln L(\theta) = 5 \ln \theta - 70 \theta \\ &\hookrightarrow \ln \text{ strictly increasing (max } L \text{ max } \ln L) \\ \frac{d}{d\theta} (\theta^5 e^{-70 \theta}) &= 5 \theta^4 e^{-70 \theta} - 70 \theta^5 e^{-70 \theta} \\ 0 &= (5 - 70 \theta) \theta^4 e^{-70 \theta} \end{aligned}$$

$$\therefore \hat{\theta} = \frac{5}{70} = \frac{1}{14}$$

$$\begin{aligned} \frac{d}{d\theta} \ln(\theta^5 e^{-70 \theta}) &= \frac{d}{d\theta} \ln \theta^5 \\ &\hookrightarrow \ln(ab) = \ln a + \ln b \quad ; \quad \ln(a^b) = b \ln a \\ &\hookrightarrow \ln(\theta^5 e^{-70 \theta}) \\ &= \ln(\theta^5) + \ln(e^{-70 \theta}) \\ &= 5 \ln \theta - 70 \theta \end{aligned}$$

$$\frac{d}{d\theta} (5 \ln \theta - 70 \theta) = \frac{5}{\theta} - 70 = 0$$

$$\theta = \frac{5}{70} = \frac{1}{14}$$

3) box k balls $\bar{k} \Rightarrow \text{white} \quad \{ > 0 \}$
 $k - \bar{k} \Rightarrow \text{red} \quad \{ = 0 \}$

a)

$X_i \rightarrow$ number on i th ball drawn

$Y_i \rightarrow$ number on i th white ball

Work through memo

Prove $E[\hat{S}] = E[\bar{S}] = E[\tilde{S}] = 0$

$S \rightarrow$ sum of all ball numbers

$$E[X_i] = \frac{S}{k} \quad \cancel{L} \quad L = \frac{\bar{S}}{k} = \frac{1}{N} \sum_{i=1}^n X_i = \frac{1}{N} \sum_{j=1}^{\bar{N}} Y_j$$

$$E[Y_i] = \frac{S}{k}$$

$$E[\hat{S}] = E\left[\frac{k}{n} \sum_{i=1}^n X_i\right] = \frac{k}{n} \sum_{i=1}^n E[X_i] = \frac{k}{n} \sum_{i=1}^n \left(\frac{S}{k}\right)$$

$$= S \quad \checkmark$$

$$\bar{N} = n - (n - \bar{k}) = \bar{k}$$

$$E[\bar{S}] = E\left[\frac{\bar{k}}{\bar{N}} \sum_{i=1}^{\bar{N}} X_i\right] = \frac{\bar{k}}{\bar{N}} \sum_{i=1}^{\bar{N}} E[X_i] = \frac{\bar{k}}{n - (n - \bar{k})} \sum_{i=1}^{\bar{N}} \left(\frac{S}{k}\right)$$

$$= \frac{\bar{k}}{\bar{N}} \sum_{i=1}^{\bar{N}} \frac{S}{k}$$

$$E[\tilde{S}] = E\left[\frac{\bar{k}}{m} \sum_{i=1}^m Y_i\right] = \frac{\bar{k}}{m} \sum_{i=1}^m E[Y_i]$$

$$= \frac{\bar{k}}{m} \sum_{i=1}^m \frac{S}{k} = S \quad \checkmark$$

$$E[L] = \sum_{i=1}^n E[L | \bar{N} = \bar{n}] p_{\bar{N}}(\bar{n})$$

$$\checkmark \quad \bar{S} = \frac{\bar{k}}{\bar{N}} \sum_{i=1}^{\bar{N}} X_i$$

$\bar{N} \rightarrow$ random / not constant

$$\hookrightarrow E\left[\frac{A}{N}\right] \neq \frac{E[A]}{E[N]}$$

$$\circ \circ \quad L = \frac{\bar{S}}{k}$$

$$L = \frac{1}{N} \sum_{i=1}^n X_i = \frac{1}{N} \sum_{i=1}^{\bar{N}} Y_i$$

(sum all or white equal as $S_{\text{red}} = 0$)

So sum over all $N = v$

$$E[L | N = v] = E\left[\frac{1}{v} \sum_{j=1}^v Y_j | N = v\right] = \frac{1}{v} \cdot v \cdot E[Y_i] = \frac{S}{k}$$

$$\circ \circ \quad E[\bar{S}] = \bar{k} \cdot \frac{S}{k} = S \quad \checkmark$$

b) $\text{var}(\tilde{S})$; $\text{var}(\hat{S})$

$$\text{var}(\tilde{S}) = \text{var}\left(\frac{\bar{k}}{m} \sum_{i=1}^m Y_i\right) = \frac{\bar{k}}{m} \cdot m \cdot \text{var}(Y_i) = \bar{k} \cdot \text{var}(Y_i)$$

$$\text{var}(Y_i) = \frac{(b-a)(b-a+2)}{12} = \frac{(\bar{k})(\bar{k}+2)}{12} = \frac{\bar{k}^2 + 2\bar{k}}{12}$$

~~$$\text{var}(\hat{S}) = \text{var}\left(\frac{k}{n} \sum_{i=1}^n X_i\right) = \frac{k}{n} \cdot n \cdot \text{var}(X_i) = k \cdot \text{var}(X_i)$$~~

$$\text{var}(\hat{S}) = \text{var}\left(\frac{k}{n} \sum_{i=1}^n X_i\right) = \left(\frac{k}{n}\right)^2 \cdot n \cdot \text{var}(X_i) = \frac{k^2}{n} \text{var}(X_i)$$

$$\text{var}(\tilde{S}) = \text{var}\left(\frac{\bar{k}}{m} \sum_{i=1}^m Y_i\right) = \left(\frac{\bar{k}}{m}\right)^2 \cdot m \cdot \text{var}(Y_i) = \frac{\bar{k}^2}{m} \text{var}(Y_i)$$

↳ variance scaling.

↓ Walkthrough

$$\text{var}(\hat{S}) = \frac{k^2}{n} \text{var}(X_i)$$

$$= \frac{k^2}{n} \left[E[X_i^2] - E[X_i]^2 \right]$$

~~$$= \frac{k^2}{n} \left[\frac{1}{k} \sum_i X_i^2 \cdot \left(\frac{1}{k}\right) - \left(\frac{S}{k}\right)^2 \right]$$~~

~~$$= \frac{k^2}{n} \left[\frac{1}{k} - \frac{S^2}{k^2} \right] = \frac{k^2}{n} \left[\frac{k - S^2}{k^2} \right]$$~~

$$E[X_i] = \rho E[Y_i] \quad E[X_i^2] = \rho E[Y_i^2]$$

$$= \frac{k^2}{n} \left[\rho E[Y_i^2] - \rho^2 (E[Y_i])^2 \right]$$

$$= \frac{k^2}{n} \left[\rho E[Y_i^2] - (E[Y_i])^2 \right]$$

$$= \frac{k^2}{n} \left[\text{var}(Y_i) + \rho E[Y_i^2] - E[Y_i]^2 + E[Y_i^2] - (E[Y_i])^2 \right]$$

$$= \frac{k^2}{n} \left[\text{var}(Y_i) + E[Y_i^2] \left(\frac{1}{\rho} - 1\right) \right]$$

$$= \frac{\bar{k}^2}{n} \text{var}(Y_i) \left[1 + \frac{n(1-\rho)}{\rho} \right]$$

$$= \frac{\bar{k}^2}{m} \cdot \frac{m}{n} \cdot \text{var}(Y_i) \left[\frac{\rho + n(1-\rho)}{\rho} \right]$$

$$= \text{var}(\tilde{S}) \cdot \frac{m}{n} \cdot \frac{\rho + n(1-\rho)}{\rho}$$

$$\circ \circ \text{ var}(\hat{S}) = \text{var}(\bar{S}) \cdot \frac{m}{n} \cdot \frac{p+r(1-p)}{p}$$

$$\hookrightarrow \circ \circ \frac{\text{var}(\hat{S})}{\text{var}(\bar{S})} = \frac{\overbrace{m}^{m=n}}{m} \cdot \frac{p+r(1-p)}{p}$$

$$\hookrightarrow \frac{\text{var}(\hat{S})}{\text{var}(\bar{S})} = \frac{p}{p+r(1-p)} \quad \checkmark$$

$$\hookrightarrow \circ \circ \text{ var}(\hat{S}) \approx \text{var}(\bar{S})$$

$$\text{to } \circ \circ \frac{m}{n} \cdot \frac{p+r(1-p)}{p} \approx 1$$

$$n \approx \frac{np}{p+r(1-p)} \quad \checkmark$$

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